

Using all previous investigation findings, create a professional SOC Incident Report. Include:

- Executive Summary
- Detection Summary
- Timeline
- Investigation Findings
- Indicators of Compromise
- MITRE ATT&CK Mapping
- Severity Assessment
- Containment Actions
- Recommendations

Clearly distinguish confirmed evidence from investigative assumptions.

**SOC Incident Report**

**Incident ID:** SOC-IR-2026-0703-001  
**Analyst:** Tier-1 SOC Analyst  
**Date:** July 3, 2026  
**Status:** Escalated to Tier-2 / Incident Response  
**Incident Classification:** Suspected Credential Compromise with Malicious PowerShell Activity

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**1. Executive Summary**

A sequence of Windows Security and Sysmon events indicates a probable cyber intrusion involving the successful use of a privileged **Administrator** account. The attack began with multiple failed authentication attempts from a single internal IP address (192.168.10.25), followed by a successful login from the same source five minutes later.

Immediately after authentication, an encoded PowerShell command was executed, and the PowerShell process established an outbound HTTPS connection to the external IP address 203.0.113.50. This sequence is consistent with common post-compromise attacker behavior, including execution of obfuscated commands and potential communication with a Command-and-Control (C2) server.

While the evidence strongly suggests malicious activity, the exact purpose of the PowerShell command and the content of the outbound network traffic cannot be confirmed without additional forensic analysis.

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**2. Detection Summary**

| Event                 | Source               | Detection                                         |
|-----------------------|----------------------|---------------------------------------------------|
| Windows Event ID 4625 | Windows Security Log | Multiple failed Administrator login attempts      |
| Windows Event ID 4624 | Windows Security Log | Successful Administrator login from same IP       |
| Sysmon Event ID 1     | Sysmon               | Encoded PowerShell execution                      |
| Sysmon Event ID 3     | Sysmon               | Outbound HTTPS connection initiated by PowerShell |

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**3. Attack Timeline**

| Time                     | Event                                                                           | Assessment                                     |
|--------------------------|---------------------------------------------------------------------------------|------------------------------------------------|
| <b>T0</b>                | Multiple failed login attempts (Event ID 4625) from 192.168.10.25               | Suspected brute-force or password guessing     |
| <b>T0 + 5 min</b>        | Successful Administrator login (Event ID 4624)                                  | Possible successful use of valid credentials   |
| <b>Immediately After</b> | Encoded PowerShell command executed (Sysmon Event ID 1)                         | Suspicious execution of obfuscated commands    |
| <b>Shortly After</b>     | PowerShell established HTTPS connection to 203.0.113.50:443 (Sysmon Event ID 3) | Possible C2 communication or payload retrieval |

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## 4. Investigation Findings

### Confirmed Evidence

#### Authentication Activity

- Multiple failed login attempts targeted the **Administrator** account.
- Successful login occurred five minutes later from the same source IP (192.168.10.25).

#### Process Execution

- powershell.exe was launched using an encoded command.
- Execution occurred after the successful privileged login.

#### Network Activity

- PowerShell established an outbound HTTPS connection.
- Destination IP: 203.0.113.50
- Destination Port: 443

#### Persistence

- No evidence of persistence mechanisms (e.g., Registry Run Keys, Scheduled Tasks, Services) was observed in the provided logs.
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### Investigative Assumptions

The following conclusions are supported by the event sequence but are **not confirmed** by the available evidence:

- The successful login may have resulted from a brute-force attack or the use of stolen credentials.
- The encoded PowerShell command may have been used to execute malware, download additional payloads, or evade detection.

- The outbound HTTPS connection may represent Command-and-Control (C2), payload retrieval, beaconing, or data exfiltration.
- Additional attacker objectives, such as lateral movement or persistence, cannot be confirmed without further evidence.

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## 5. Indicators of Compromise (IoCs)

### Network IoCs

| Indicator     | Type                       |
|---------------|----------------------------|
| 192.168.10.25 | Source IP (Authentication) |
| 203.0.113.50  | External Destination IP    |
| TCP Port 443  | Outbound HTTPS Connection  |

### Host IoCs

| Indicator                                           | Type                              |
|-----------------------------------------------------|-----------------------------------|
| powershell.exe                                      | Process                           |
| Encoded PowerShell Command (-EncodedCommand / -enc) | Suspicious Command-Line Parameter |
| Administrator Account                               | Targeted Privileged Account       |

### Security Events

| Event ID          | Description        |
|-------------------|--------------------|
| 4625              | Failed Logon       |
| 4624              | Successful Logon   |
| Sysmon Event ID 1 | Process Creation   |
| Sysmon Event ID 3 | Network Connection |

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## 6. MITRE ATT&CK Mapping

| ATT&CK Tactic                    | Technique ID | Technique Name | Confidence  |
|----------------------------------|--------------|----------------|-------------|
| Credential Access                | T1110        | Brute Force    | High        |
| Initial Access / Defense Evasion | T1078        | Valid Accounts | Medium-High |

| ATT&CK Tactic                     | Technique ID | Technique Name                                | Confidence  |
|-----------------------------------|--------------|-----------------------------------------------|-------------|
| Execution                         | T1059.001    | Command and Scripting Interpreter: PowerShell | High        |
| Defense Evasion                   | T1027        | Obfuscated Files or Information               | High        |
| Command and Control               | T1071.001    | Application Layer Protocol: Web Protocols     | Medium-High |
| Command and Control<br>(Possible) | T1105        | Ingress Tool Transfer                         | Medium      |
| Persistence                       | —            | No confirmed persistence observed             | N/A         |

## 7. Severity Assessment

**Overall Severity: Critical**

### Justification

The incident is classified as **Critical** based on the following factors:

- Targeting and successful authentication of a privileged **Administrator** account.
- Multiple failed login attempts followed by a successful login from the same source IP.
- Execution of an encoded PowerShell command, a common technique for obfuscation and malicious execution.
- Outbound encrypted communication from PowerShell to an external IP address.
- Correlation across authentication, process creation, and network events indicating a likely active compromise.

## 8. Containment Actions

### Immediate Actions

1. Isolate the affected endpoint from the network.
2. Disable or reset the **Administrator** account credentials and rotate other privileged credentials if compromise is suspected.
3. Block outbound communication to 203.0.113.50 at firewalls, proxies, and endpoint controls.
4. Preserve forensic evidence, including memory, running processes, active network connections, and relevant log files.
5. Collect Windows Security, Sysmon, PowerShell, EDR, firewall, and proxy logs.
6. Escalate the incident to the Tier-2 SOC or Incident Response team for deeper forensic analysis.

## Threat Hunting

- Search for additional logons from 192.168.10.25.
  - Hunt for other systems executing encoded PowerShell commands.
  - Identify hosts communicating with 203.0.113.50.
  - Review authentication activity for other privileged accounts.
  - Check for persistence mechanisms created after the observed events.
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## 9. Recommendations

### Short-Term Recommendations

- Verify whether the successful Administrator login was authorized.
- Decode and analyze the PowerShell command.
- Perform reputation analysis on 203.0.113.50.
- Review PowerShell Operational logs and EDR telemetry.
- Scan the affected host for malware and unauthorized changes.
- Monitor for additional outbound connections to suspicious infrastructure.

### Long-Term Recommendations

- Enable Multi-Factor Authentication (MFA) for privileged accounts where possible.
  - Restrict or harden use of the built-in Administrator account.
  - Implement PowerShell Constrained Language Mode and Script Block Logging.
  - Enforce account lockout policies to mitigate brute-force attacks.
  - Deploy and tune Sigma/EDR detections for:
    - Failed authentication bursts
    - Encoded PowerShell execution
    - Registry persistence
    - Suspicious DNS queries
    - PowerShell-initiated network connections
  - Regularly review privileged account usage and conduct proactive threat hunting using the mapped MITRE ATT&CK techniques.
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## Conclusion

### Confirmed Evidence

- Multiple failed logon attempts against the **Administrator** account.
- Successful Administrator authentication from the same source IP.

- Execution of an encoded PowerShell command.
- Outbound HTTPS connection from powershell.exe to 203.0.113.50.
- No confirmed evidence of persistence in the available logs.

### **Investigative Assumptions**

- The successful login likely resulted from credential compromise or successful brute-force activity.
- The encoded PowerShell command likely executed malicious or obfuscated instructions.
- The outbound HTTPS connection may represent Command-and-Control, payload retrieval, or data exfiltration.
- The attacker's ultimate objective (e.g., persistence, lateral movement, ransomware deployment, or data theft) cannot be conclusively determined from the available evidence.

### **Final Assessment**

The correlated evidence indicates a **high-confidence suspected host compromise** involving a privileged account. The combination of suspicious authentication activity, encoded PowerShell execution, and external network communication represents a credible threat requiring **immediate containment, forensic investigation, and escalation to the Incident Response team.**